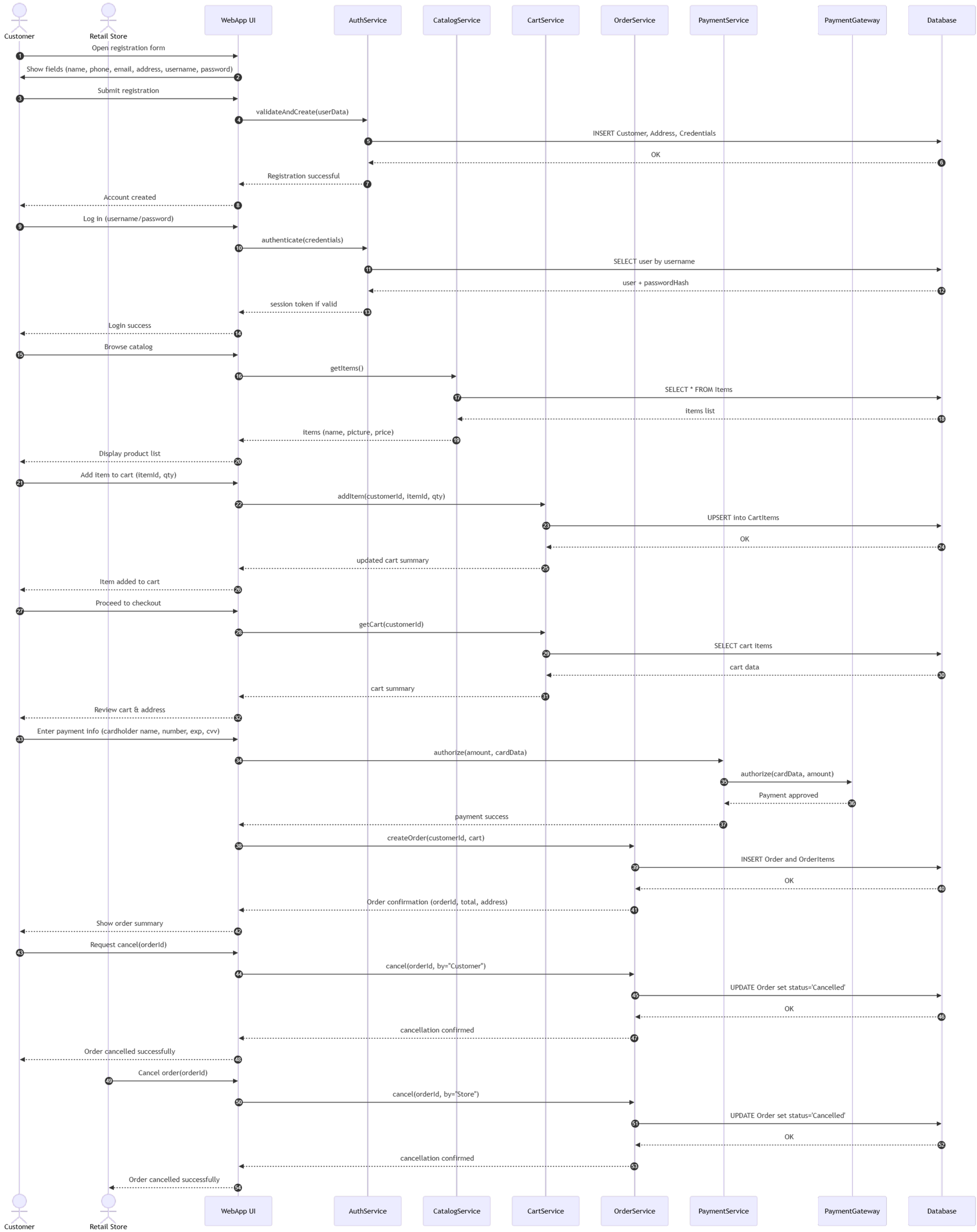
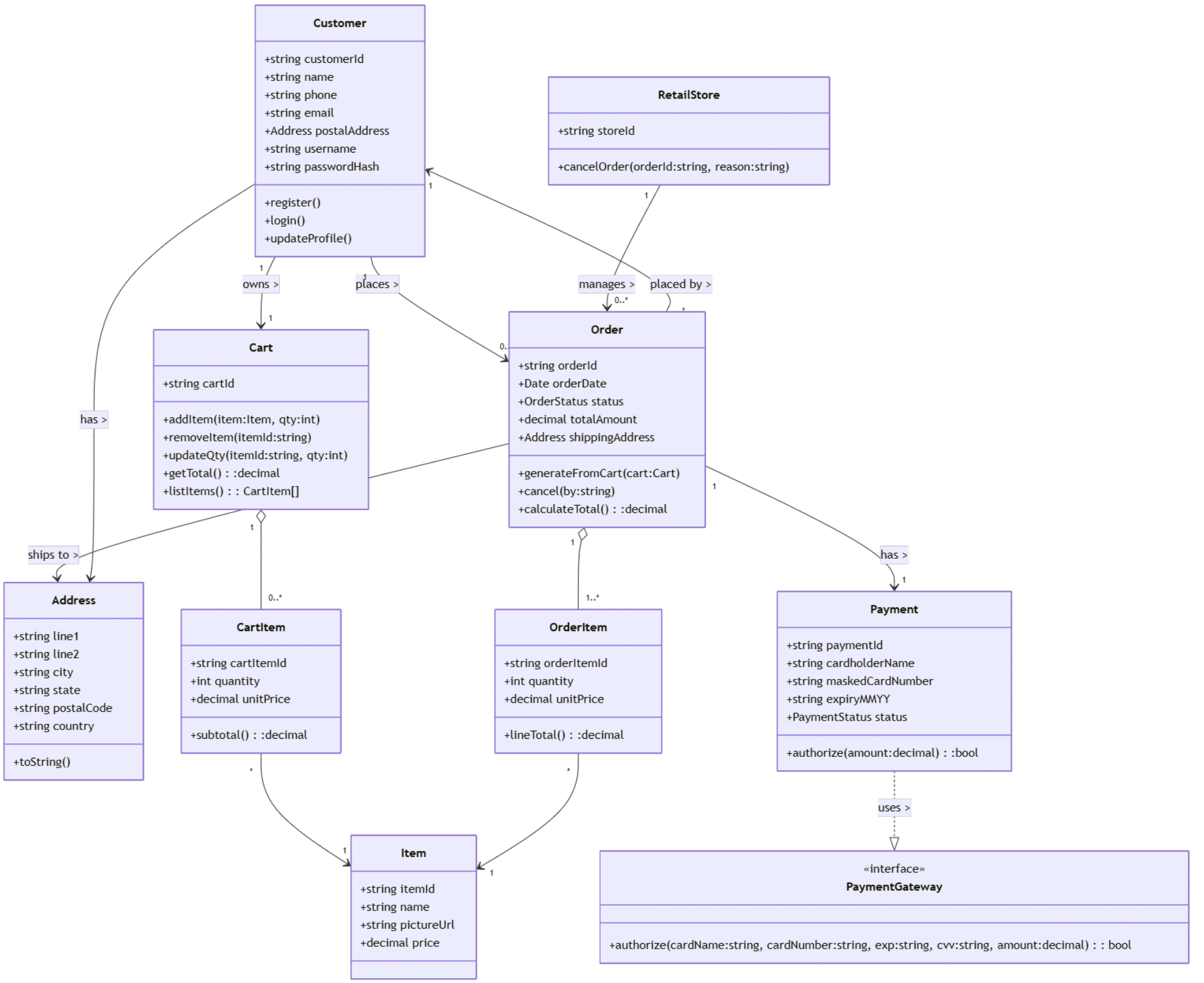
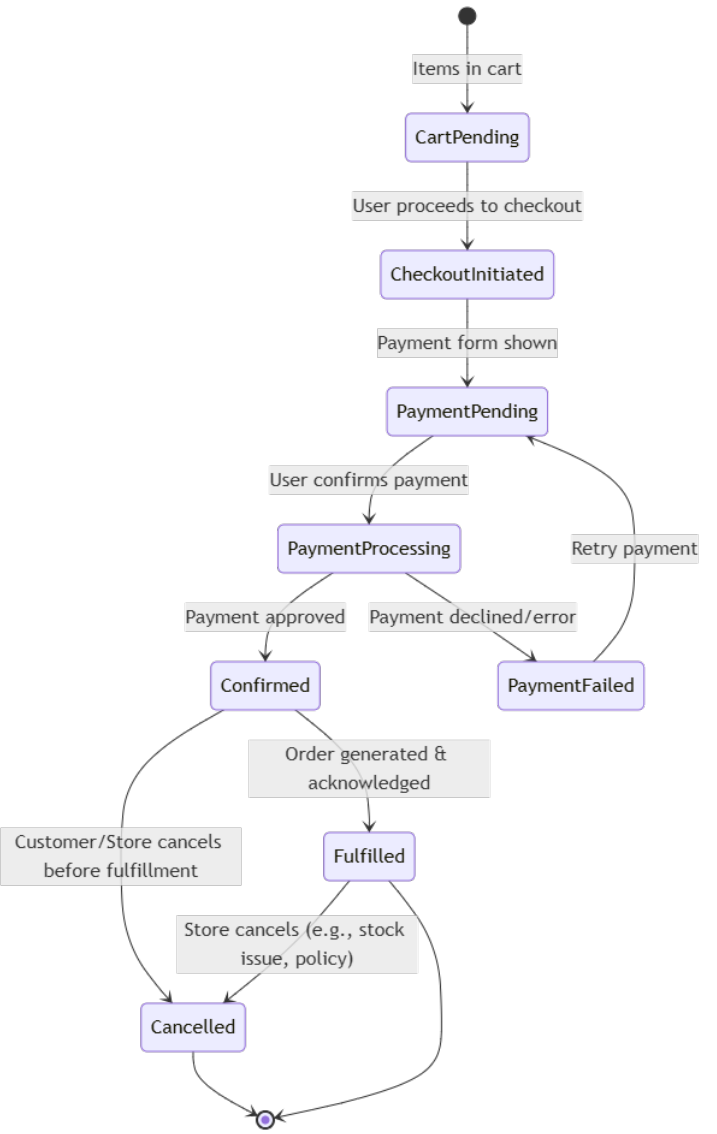
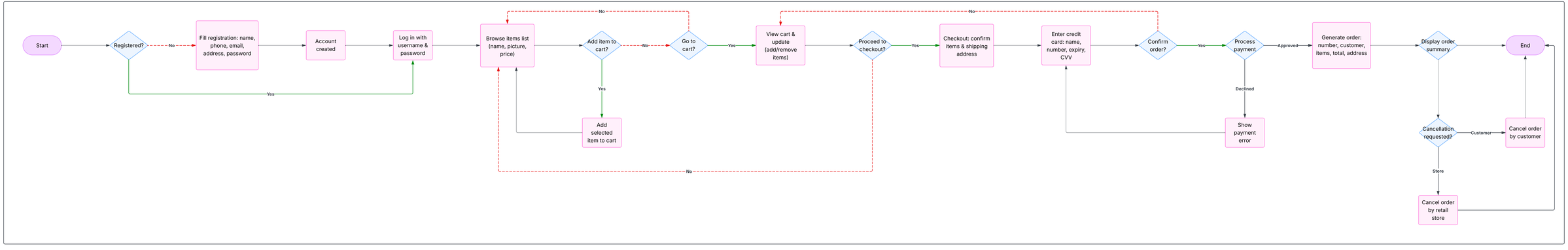
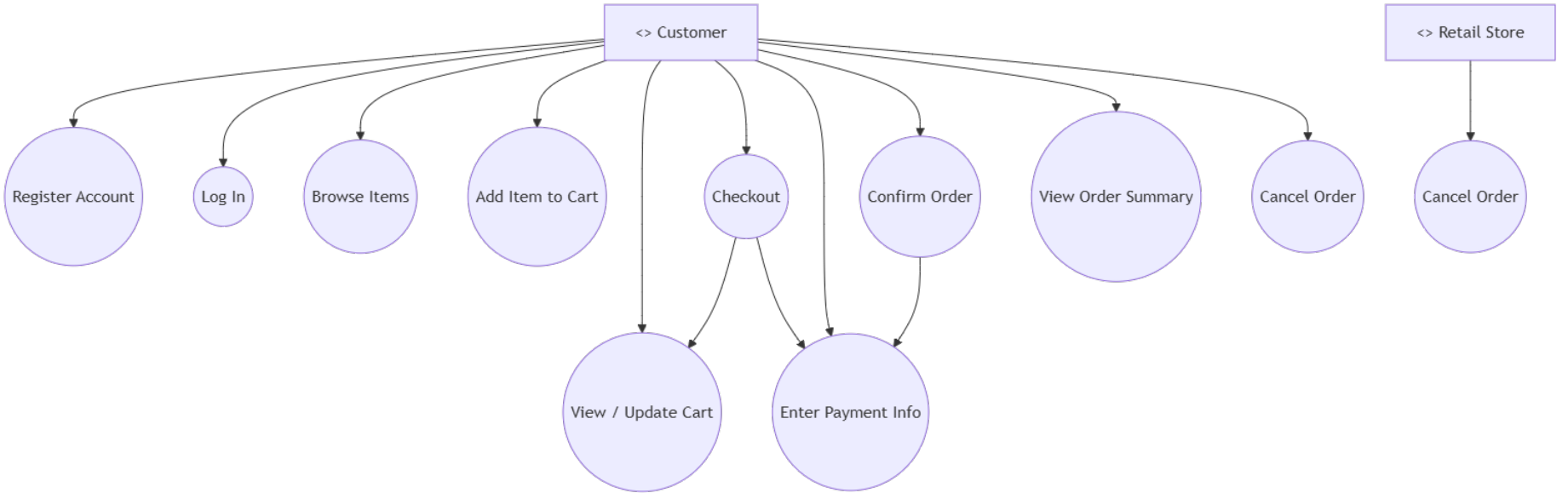

Customer
Retail Store
WebApp UI
AuthService
CatalogService
CartService
OrderService
PaymentService
PaymentGateway
Database
1 Open registration form
2 Show fields (name, phone, email, address, username, password)
3 Submit registration
4 validateAndCreate(userData)
5 INSERT Customer, Address, Credentials
6 OK
7 Registration successful
8 Account created
9 Log in (username/password)
10 authenticate(credentials)
11 SELECT user by username
12 user + passwordHash
13 session token if valid
14 Login success
15 Browse catalog
16 getItems()
17 SELECT * FROM Items
18 items list
19 items (name, picture, price)
20 Display product list
21 Add item to cart (itemId, qty)
22 addItem(customerId, itemId, qty)
23 UPSERT into CartItems
24 OK
25 updated cart summary
26 Item added to cart
27 Proceed to checkout
28 getCart(customerId)
29 SELECT cart items
30 cart data
31 cart summary
32 Review cart & address
33 Enter payment info (cardholder name, number, exp, cvv)
34 authorize(amount, cardData)
35 authorize(cardData, amount)
36 Payment approved
37 payment success
38 createOrder(customerId, cart)
39 INSERT Order and OrderItems
40 OK
41 Order confirmation (orderId, total, address)
42 Show order summary
43 Request cancel(orderId)
44 cancel(orderId, by="Customer")
45 UPDATE Order set status='Cancelled'
46 OK
47 cancellation confirmed
48 Order cancelled successfully
49 Cancel order(orderId)
50 cancel(orderId, by="Store")
51 UPDATE Order set status='Cancelled'
52 OK
53 cancellation confirmed
54 Order cancelled successfully


Customer
+string customerId
+string name
+string phone
+string email
+Address postalAddress
+string username
+string passwordHash
+register()
+login()
+updateProfile()
RetailStore
+string storeId
+cancelOrder(orderId:string, reason:string)
owns >
places >
manages >
placed by >
has >
ships to >
has >
uses >
Cart
+string cartId
+addItem(item:Item, qty:int)
+removeItem(itemId:string)
+updateQty(itemId:string, qty:int)
+getTotal() : :decimal
+listItems() : : CartItem[]
Order
+string orderId
+Date orderDate
+OrderStatus status
+decimal totalAmount
+Address shippingAddress
+generateFromCart(cart:Cart)
+cancel(by:string)
+calculateTotal() : :decimal
Address
+string line1
+string line2
+string city
+string state
+string postalCode
+string country
+toString()
CartItem
+string cartItemId
+int quantity
+decimal unitPrice
+subtotal() : :decimal
OrderItem
+string orderItemId
+int quantity
+decimal unitPrice
+lineTotal() : :decimal
Payment
+string paymentId
+string cardholderName
+string maskedCardNumber
+string expiryMMYY
+PaymentStatus status
+authorize(amount:decimal) : :bool
Item
+string itemId
+string name
+string pictureUrl
+decimal price
«interface»
PaymentGateway
+authorize(cardName:string, cardNumber:string, exp:string, cvv:string, amount:decimal) : : bool


Items in cart
CartPending
User proceeds to checkout
CheckoutInitiated
Payment form shown
PaymentPending
User confirms payment
PaymentProcessing
Retry payment
Payment approved
Payment declined/error
Confirmed
PaymentFailed
Order generated & acknowledged
Customer/Store cancels before fulfillment
Fulfilled
Store cancels (e.g., stock issue, policy)
Cancelled

E-commerce Checkout Flowchart




<> Customer
<> Retail Store
Register Account
Log In
Browse Items
Add Item to Cart
Checkout
Confirm Order
View Order Summary
Cancel Order
Cancel Order
View / Update Cart
Enter Payment Info